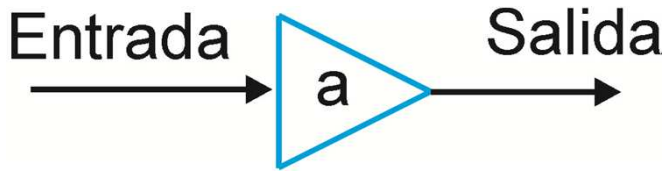
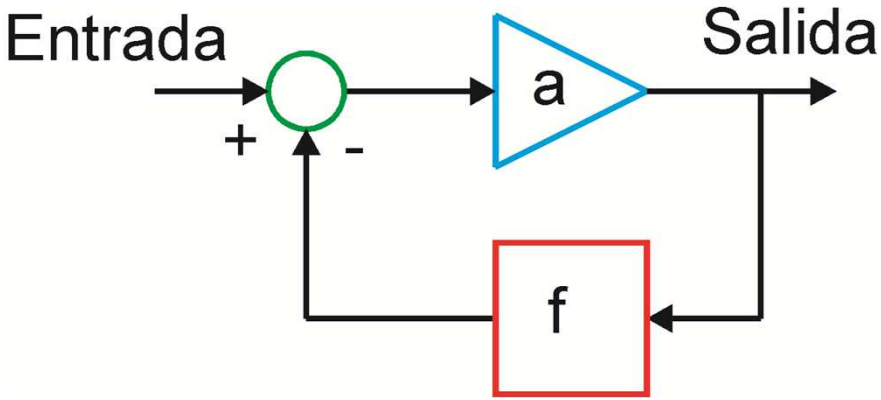
Introducción

La mayor parte de los sistemas físicos contienen alguna forma de Realimentación.





Introducción

Cuando se utiliza realimentación negativa en el diseño de amplificadores se obtienen los siguientes beneficios:

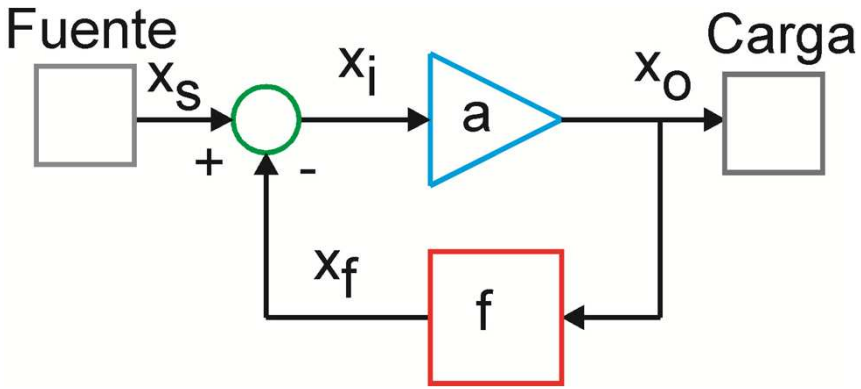
- ✓ Estabilización de la ganancia ante cambios en los parámetros del circuito.
- ✓ Reducción de la distorsión.
- ✓ Reducción del efecto del ruido (*).
- ✓ Control de las impedancias de entrada y salida.
- ✓ Ampliación del ancho de banda.

Introducción

A expensas de:

- ✓ Reducción de la ganancia.
- ✓ Mayor complejidad circuital.
- ✓ Tendencia a la inestabilidad.

Estructura General de Realimentación



$$x_o = a\,x_i$$
$$x_f = f\,x_o$$
$$x_i = x_s - x_f$$
$$A = \frac{x_o}{x_s} = \frac{a}{1 + af}$$

Ganancia de lazo: $T = af$

Si $T \gg 1 \rightarrow A \cong \frac{1}{f}$

En amplificadores las señales pueden ser corrientes o tensiones

Insensibilización de la Ganancia

$$A = \frac{a}{1 + af}$$

$$dA = \frac{da}{(1 + af)^2} \quad ; \quad \frac{dA}{A} = \frac{1}{(1 + af)} \frac{da}{a}$$

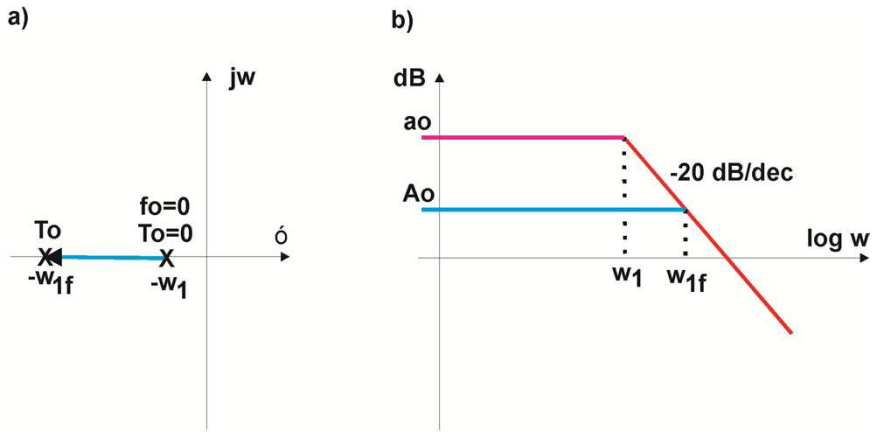
a suele tener variaciones importantes en sus parámetros.

Si T >> 1, A ≈ 1/f , y dependerá de la red de realimentación f

Esta expresión indica que el cambio relativo de **A** debido a variaciones de los parámetros del circuito, es menor que el cambio Relativo de **a**, en una cantidad **(1 + af)**

Efectos sobre el ancho de banda

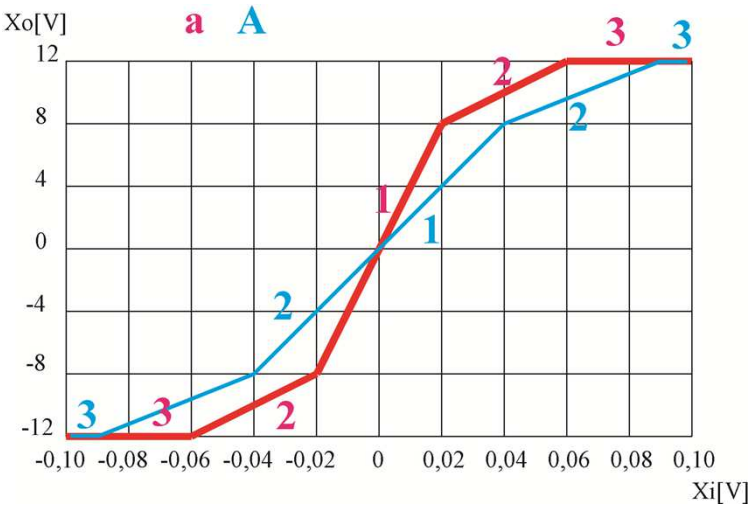
$$a(s) = \frac{a_o}{\left(1 + \frac{s}{w_1}\right)} \quad ; \quad f(s) = f_o$$



$$A(s) = \frac{a(s)}{(1 + a(s)f(s))} = \frac{A_o}{\left(1 + \frac{s}{w_{1f}}\right)}$$

$$con\colon\ T_o = a_o f_o\ ;\ \ A_o = \frac{a_o}{(1 + T_o)}\ y\ \ w_{1f} = w_1(1 + T_o)$$

Efectos sobre la distorsión



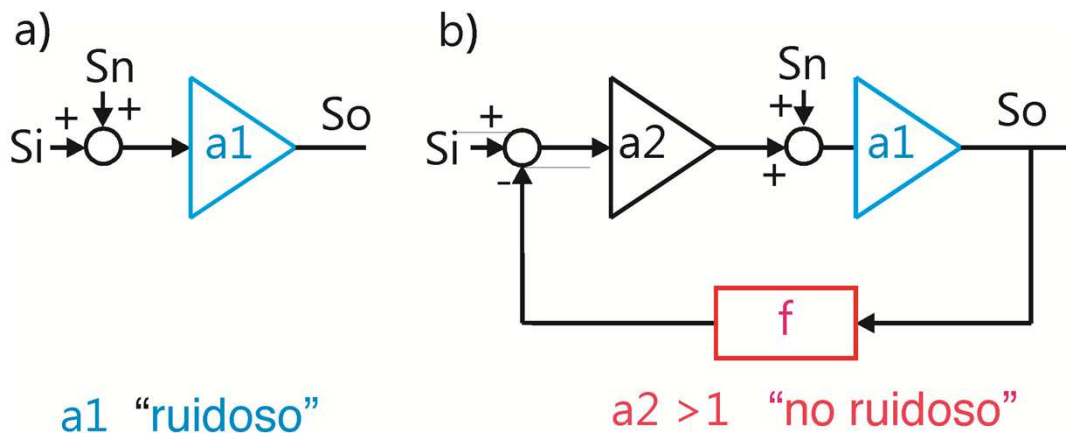
$$a_1 = 400\ ;\quad a_2 = 100\ ;\quad a_3 = 0$$

Si $f = \frac{1}{400} \rightarrow T_1 = 1\ ;\ T_2 = 0{,}25\ y\ T_3 = 0$

$A_1 = 200\ ;\quad A_2 = 80\quad y\ A_3 = 0$ (se reduce la distorsión)

Si $T_{1,2} \gg 1 \rightarrow A_{1,2} \cong {}^{1}/_{f}$; se reduce considerablemente

Efectos sobre el ruido

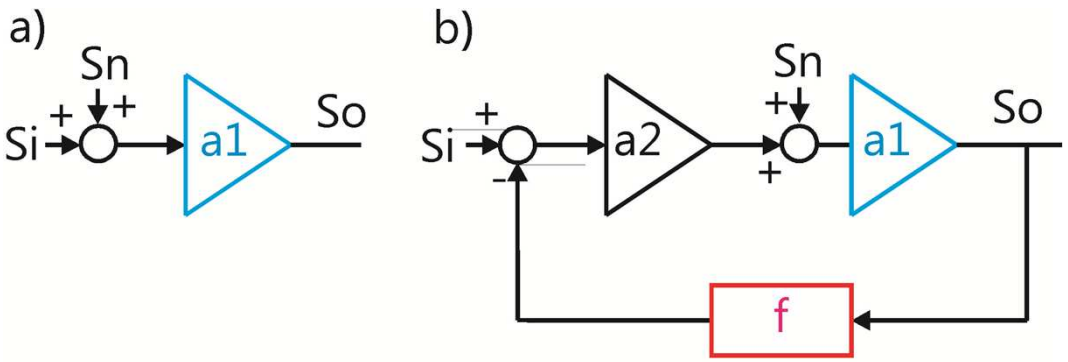


En b), se diseña **f** de modo que la ganancia sea la misma que en el caso a) con a2 > 1.

$$a)\ S_0 = a_1 S_i + a_1 S_n \rightarrow \frac{S}{N} = \frac{S_i}{S_n}$$

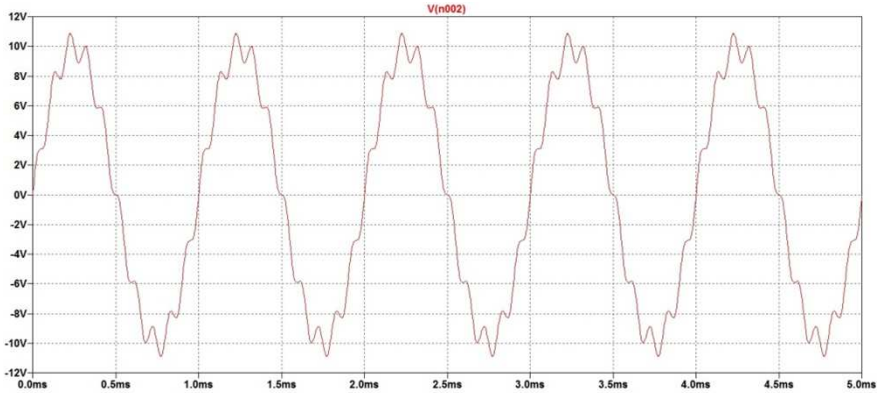
$$b)\ S_o = S_i \frac{a_1 a_2}{(1 + a_1 a_2 f)} + S_n \frac{a_1}{(1 + a_1 a_2 f)} \rightarrow \frac{S}{N} = a_2 \frac{S_i}{S_n}$$

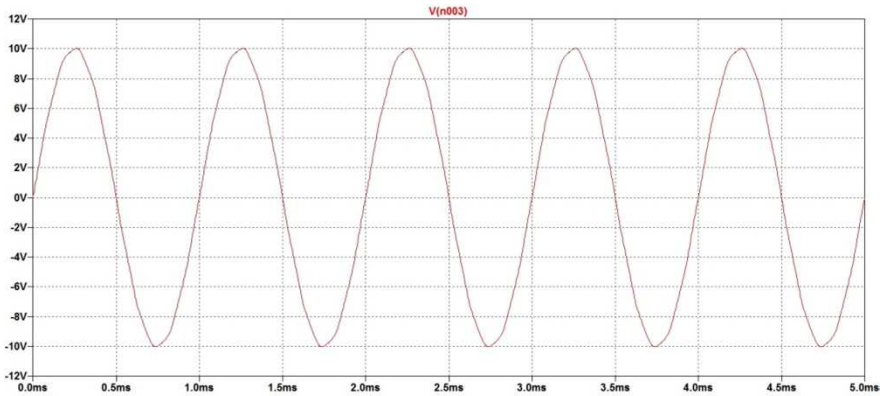
Efectos sobre el ruido



a1 "ruidoso"

a2 >1 "no ruidoso"



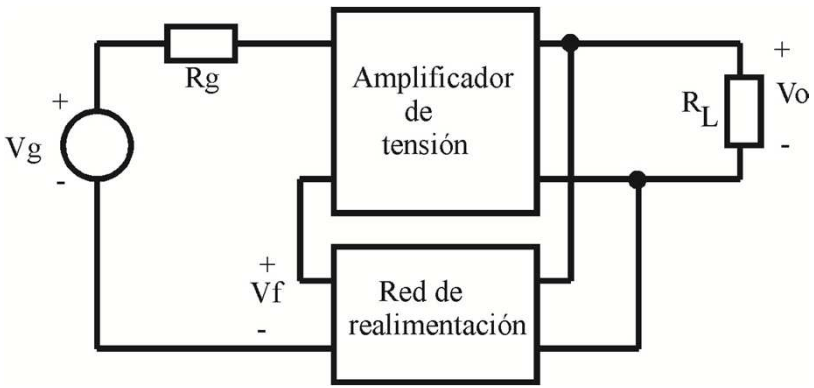


$$S_i = 1V \sin 2\pi 1000t; \quad S_n = 0.1V \sin 2\pi 10000t; \quad a_1 = 10; \quad a_2 = 10; \quad f = 0.09$$

$$caso\ a) \quad \frac{S}{N} = \frac{S_i}{S_n} = \frac{10}{1} = 10 \ \ (20dB) \quad ; \quad caso\ b) \quad \frac{S}{N} = a_2 \frac{S_i}{S_n} = 100 \ \ (40dB)$$

Configuraciones: Serie-Paralelo

Amplificadores de tensión

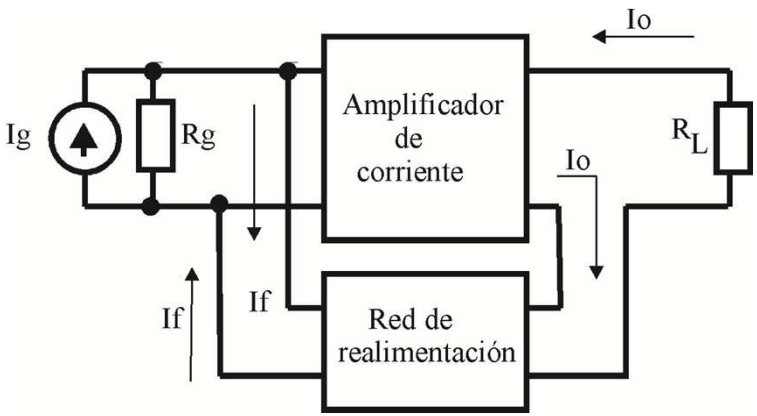


$$A_V = \frac{V_o}{V_g}$$

- ✓ Fuentes de tensión controladas por tensión.
- ✓ Alta impedancia de entrada.
- ✓ Baja impedancia de salida.
- ✓ Parámetros: [h]

Configuraciones: Paralelo-Serie

Amplificadores de corriente

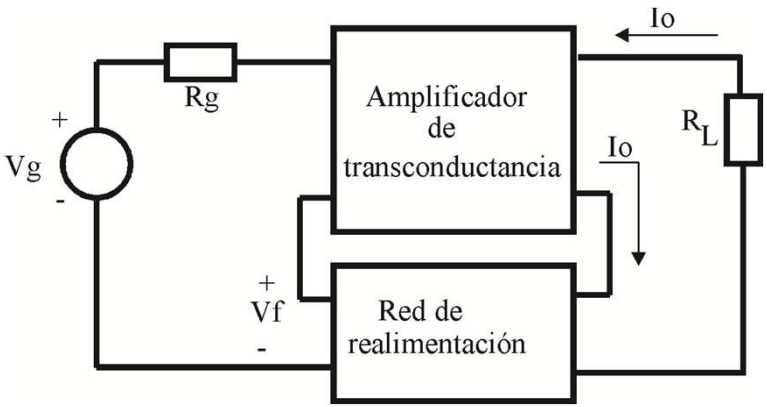


$$A_i = \frac{I_o}{I_g}$$

- ✓ Fuentes de corriente controladas por corriente.
- ✓ Baja impedancia de entrada.
- ✓ Alta impedancia de salida.
- ✓ Parámetros: [g]

Configuraciones: Serie-Serie

Amplificadores de transconductancia

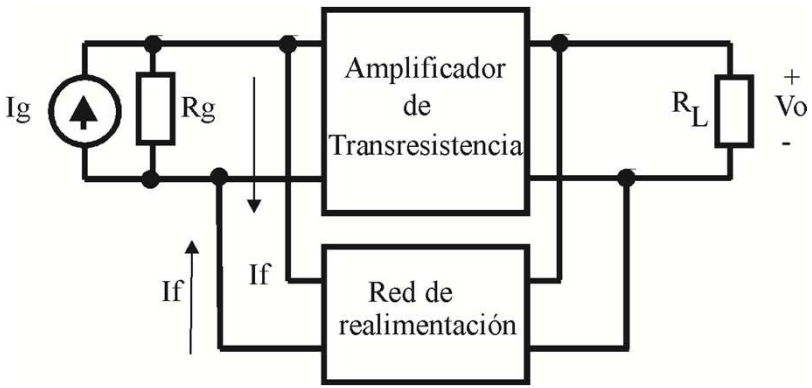


$$A_Y = \frac{I_o}{V_g}$$

- ✓ Fuentes de corriente controladas por tensión.
- ✓ Alta impedancia de entrada.
- ✓ Alta impedancia de salida.
- ✓ Parámetros: [z]

Configuraciones: Paralelo-Paralelo

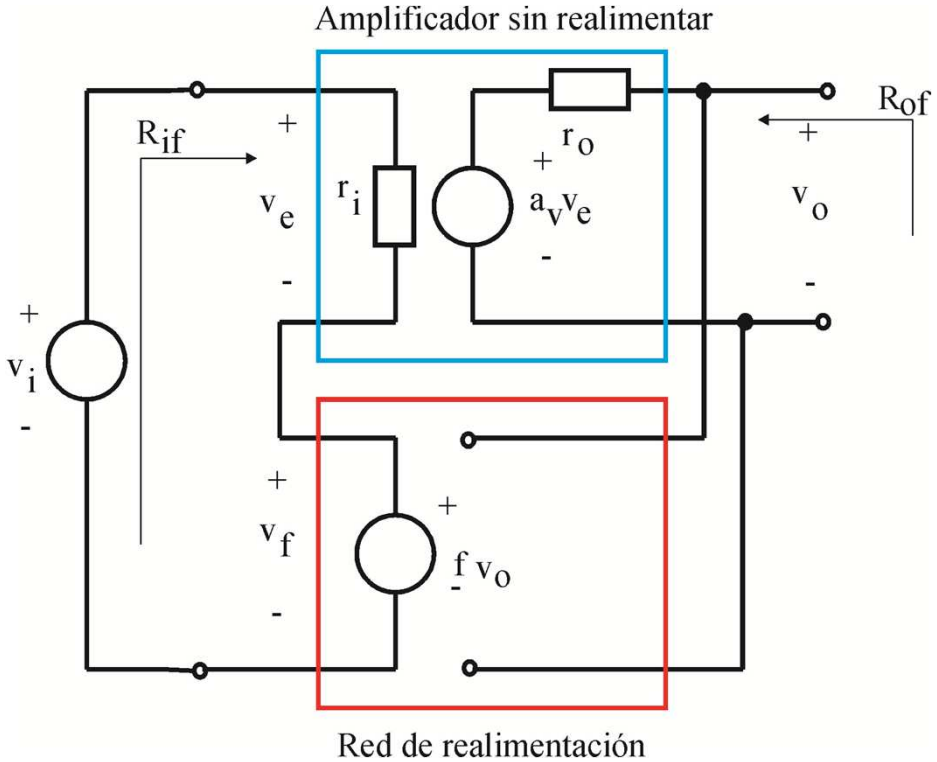
Amplificadores de transresistencia



$$A_Z = \frac{V_o}{I_g}$$

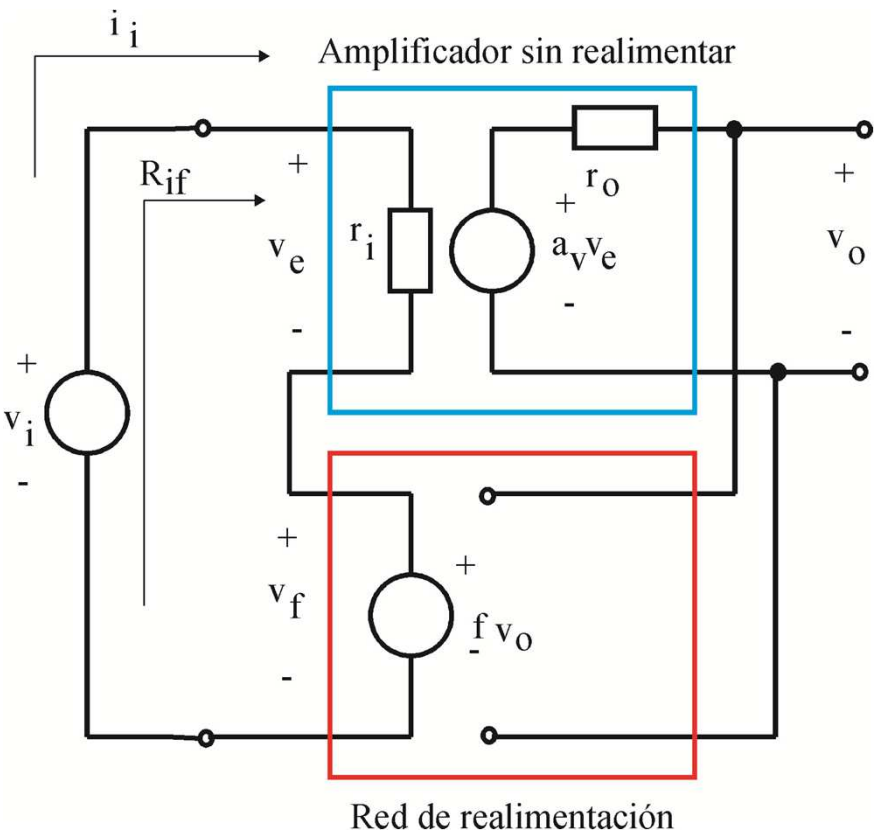
- ✓ Fuentes de tensión controladas por corriente.
- ✓ Baja impedancia de entrada.
- ✓ Baja impedancia de salida.
- ✓ Parámetros: [y]

Serie-Paralelo. Esquema ideal



$$A_V = \frac{v_o}{v_i} = \frac{a_v}{(1 + a_v f)}$$

Serie-Paralelo. Esquema ideal. Impedancia de entrada

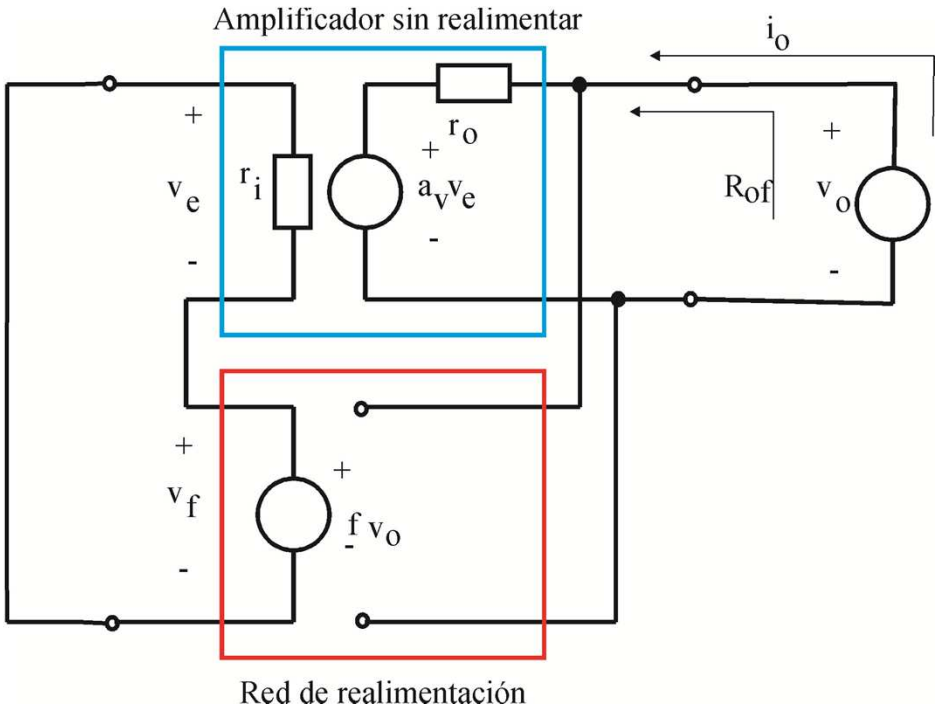


$$R_{if} = \frac{v_i}{i_i}$$

$$R_{if} = \frac{v_e + v_f}{i_i} = \frac{i_i r_i + f v_0}{i_i} = \frac{i_i r_i + f a_v i_i r_i}{i_i}$$

$$\boxed{R_{if} = r_i(1 + f a_v)}$$

Serie-Paralelo. Esquema ideal. Impedancia de salida



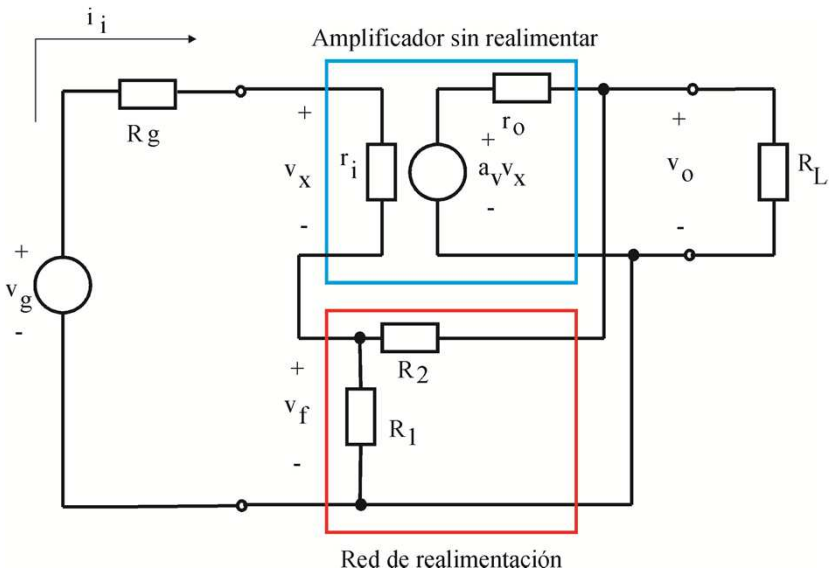
$$v_o = i_o r_o + a_v v_e$$

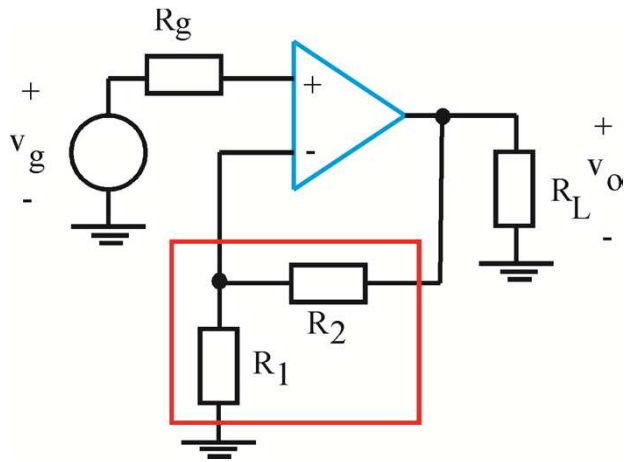
$$v_o = i_o r_o + a_v(-f v_o)$$

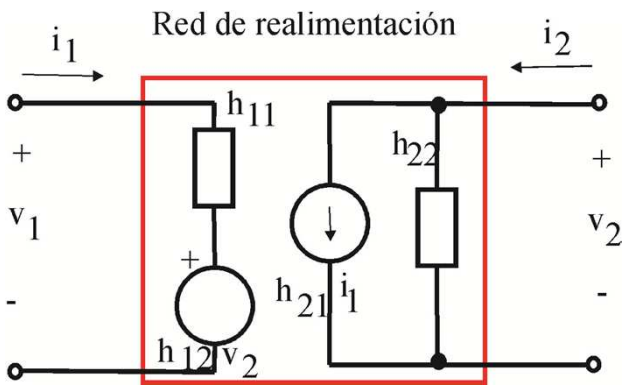
$$R_{of} = \frac{v_o}{i_o}$$

$$\boxed{R_{of} = \frac{r_o}{(1 + f a_v)}}$$

Serie-Paralelo. Esquema real



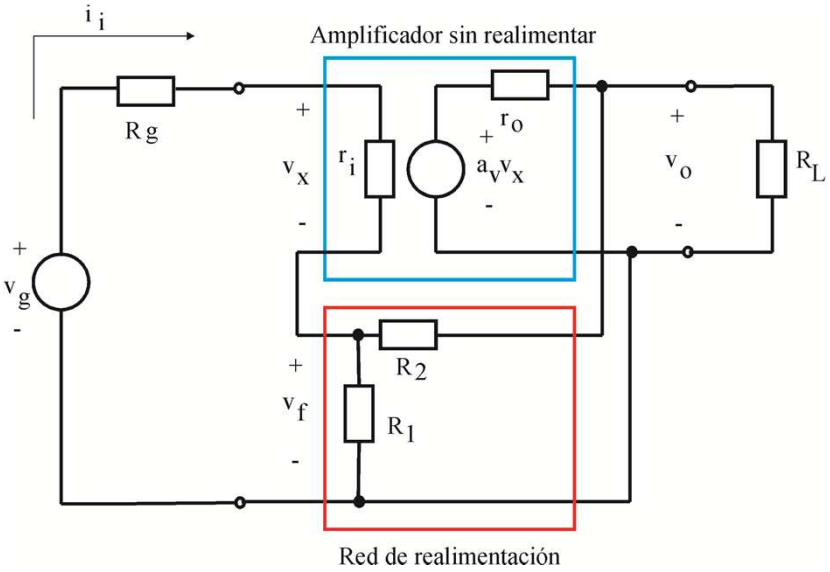


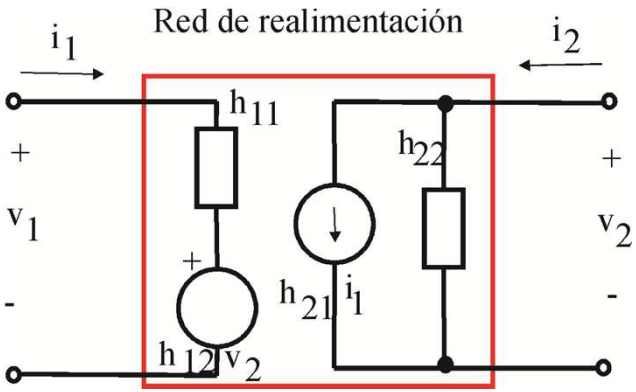


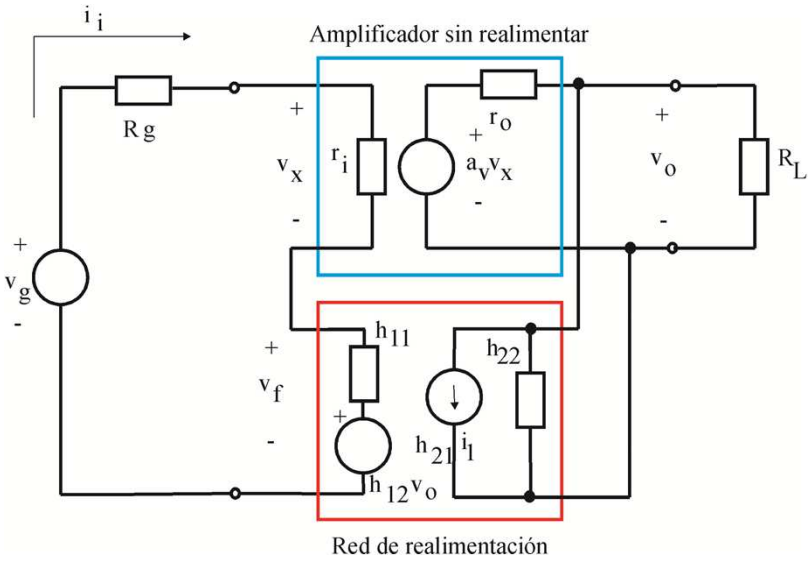
$$h_{11} = \left.\frac{v_1}{i_1}\right|_{v_2=0} = \frac{R_1 R_2}{(R_1 + R_2)} \quad ; \quad h_{12} = \left.\frac{v_1}{v_2}\right|_{i_1=0} = \frac{R_1}{(R_1 + R_2)}$$

$$h_{21} = \left.\frac{i_2}{i_1}\right|_{v_2=0} = -\frac{R_1}{(R_1 + R_2)} \quad ; \quad h_{22} = \left.\frac{i_2}{v_2}\right|_{i_1=0} = \frac{1}{(R_1 + R_2)}$$

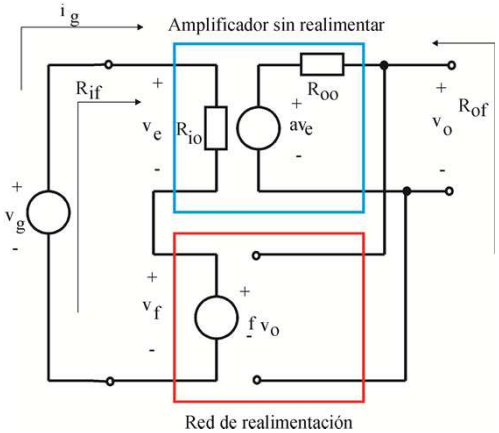
Serie-Paralelo. Esquema real







Serie-Paralelo. Esquema real



$$a = \frac{v_o}{v_g}(f = 0) = \frac{r_i}{r_i + R_g + h_{11}} a_v \frac{R_L}{R_L + r_o + r_o R_L h_{22}}$$

$$f = h_{12}$$

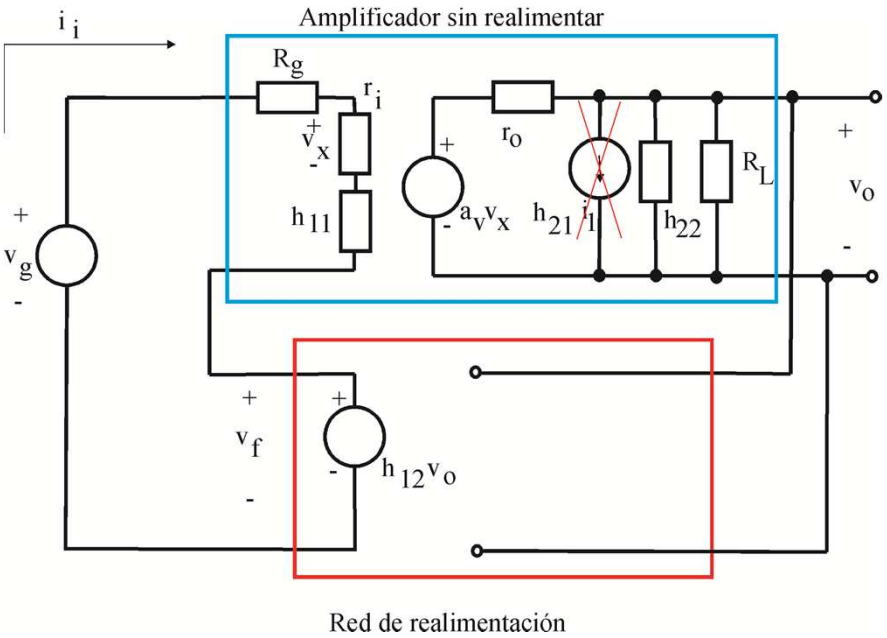
$$\boxed{A = \frac{v_o}{v_g} = \frac{a}{1 + af}}$$

$$R_{io} = R_g + r_i + h_{11}$$

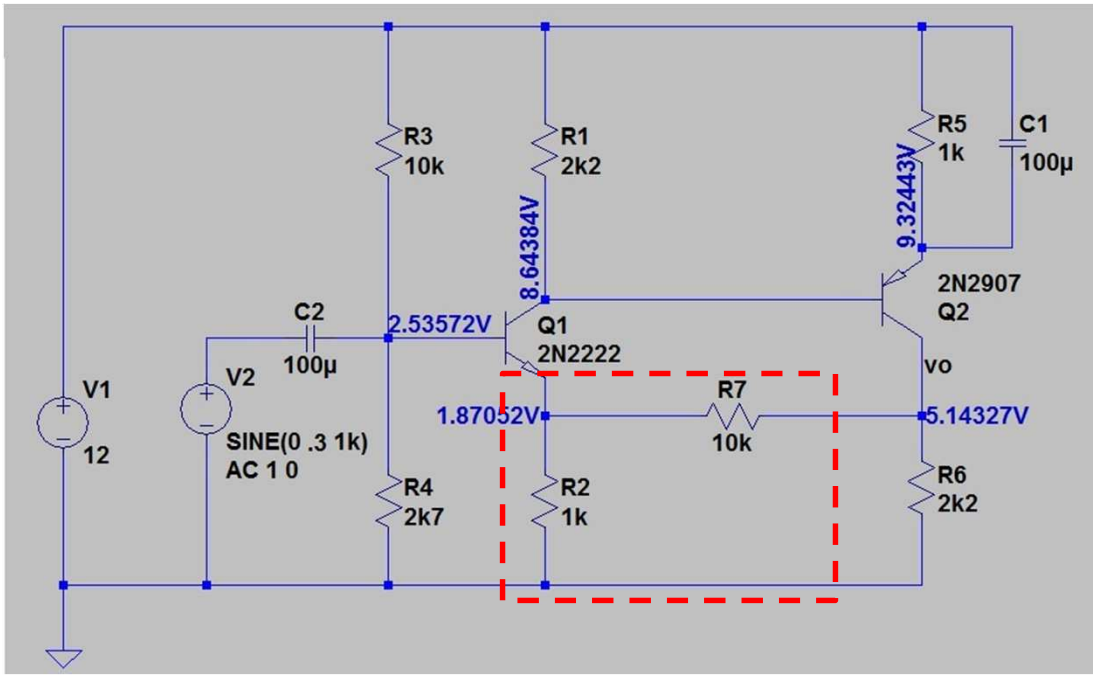
$$\boxed{R_{if} = R_{io}(1 + af)}$$

$$R_{oo} = \frac{1}{\frac{1}{R_L} + h_{22} + \frac{1}{r_o}}$$

$$\boxed{R_{of} = \frac{R_{oo}}{(1 + af)}}$$

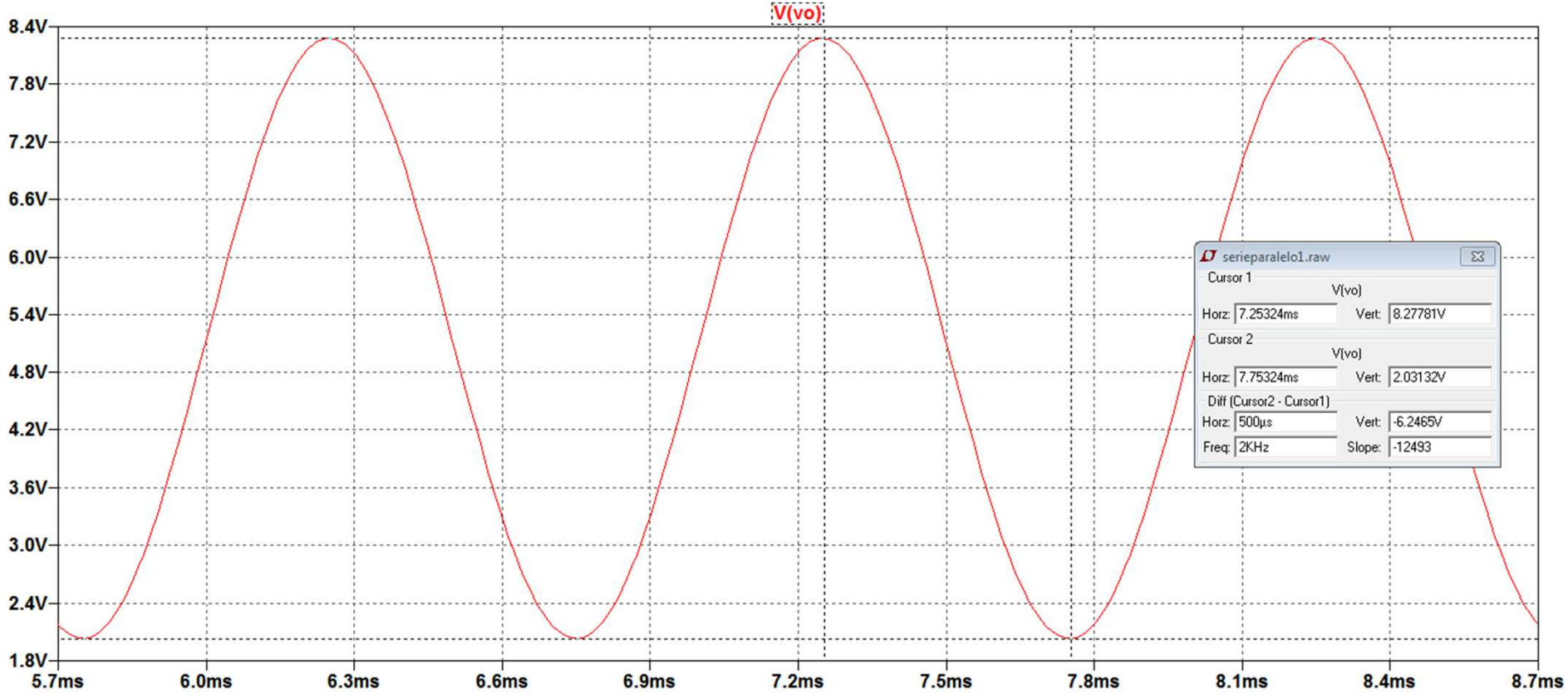


Serie-Paralelo. Ejemplo



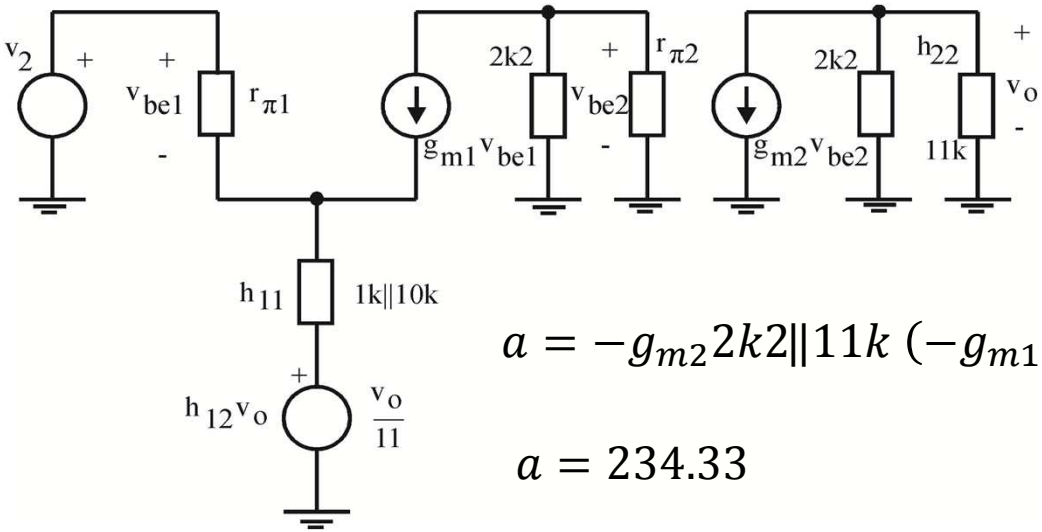
	Q1	Q2
Icq	1,54mA	2,67mA
gm	0,0591 1/Ω	0,102 1/Ω
rπ	3560 Ω	2480 Ω
hfe	210	253
Ro	69,1 K	46,3 k

Serie-Paralelo. Ejemplo



$$\frac{\widehat{\widehat{v_o}}}{\widehat{\widehat{v_2}}} = \frac{6.2465\ V}{0.6\ V} = 10.41$$

Serie-Paralelo. Ejemplo



$$a = -g_{m2}2k2\|11k\,(-g_{m1}2k2\|r_{\pi 2})\frac{r_{\pi 1}}{r_{\pi 1} + (1 + g_{m1}r_{\pi 1})1k\|10k}$$

$$a = 234.33$$

$$f = \frac{1k}{1k + 11k} = \frac{1}{11}\ ; T = af = 21.30\ ;$$

$$\boxed{A = \frac{a}{1 + T} = 10.51}$$

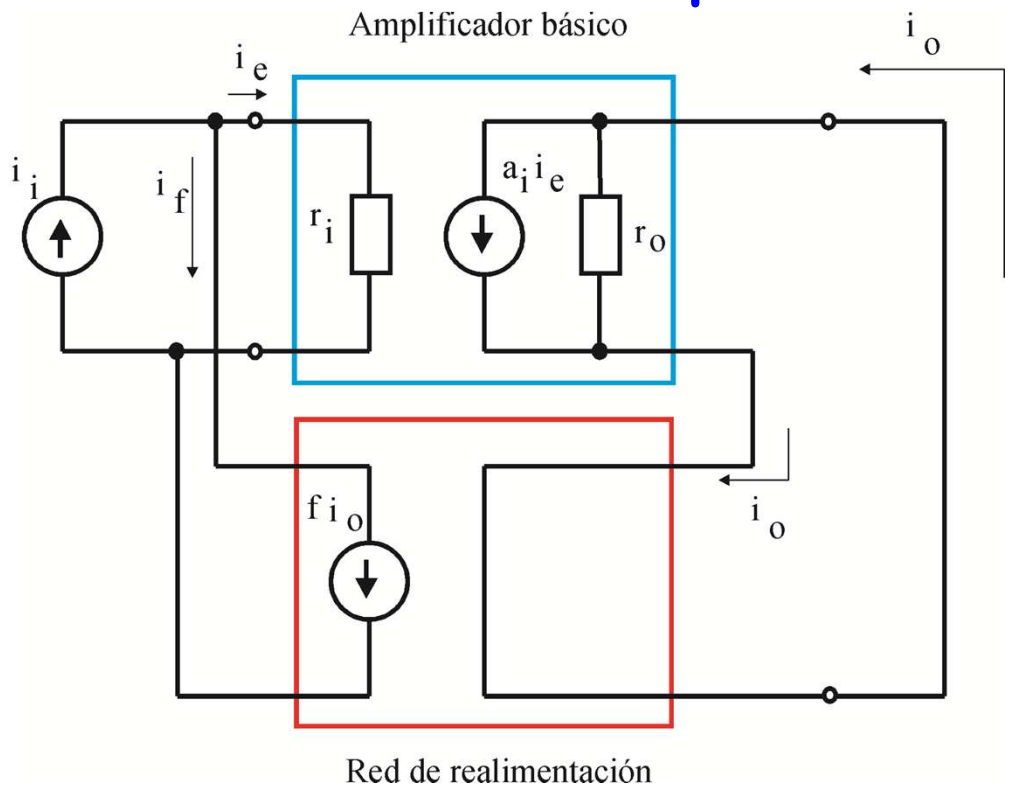
$$R_{io} = r_{\pi 1} + (1 + g_{m1}r_{\pi 1})1k\|10k = 195.74k$$

$$\boxed{R_{if} = R_{io}(1 + T) = 4.365M}$$

$$R_{oo} = 2k2\|{h_{22}}^{-1} = 1833.33\Omega$$

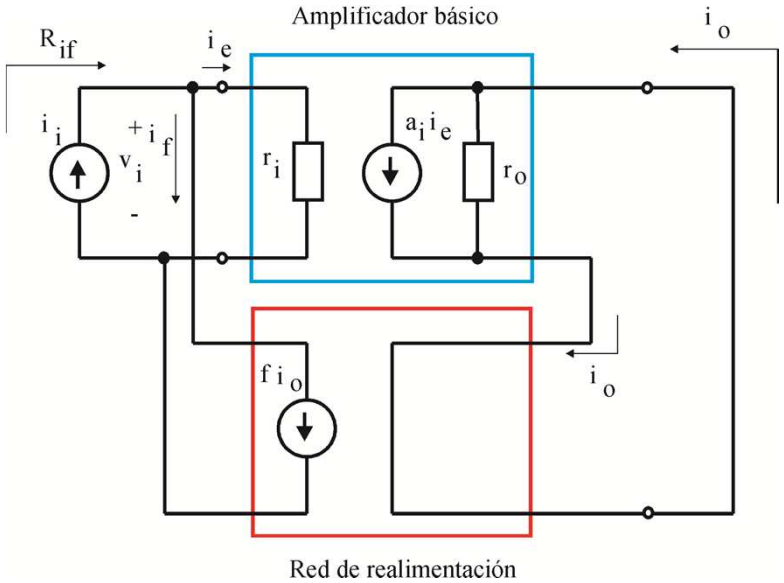
$$\boxed{R_{of} = \frac{R_{oo}}{1 + T} = 82.2\Omega}$$

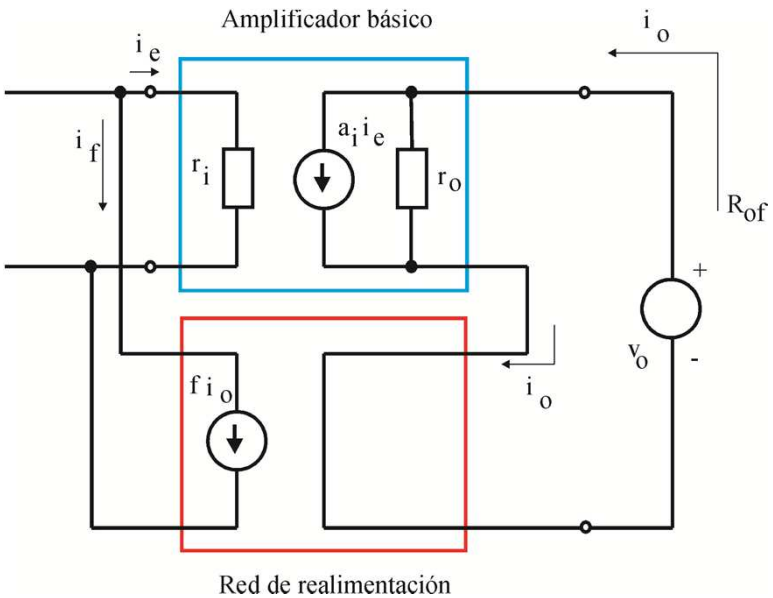
Paralelo - Serie. Esquema ideal



$$A_i = \frac{i_o}{i_i} = \frac{a_i}{(1 + a_i f)}$$

Paralelo - Serie. Esquema ideal

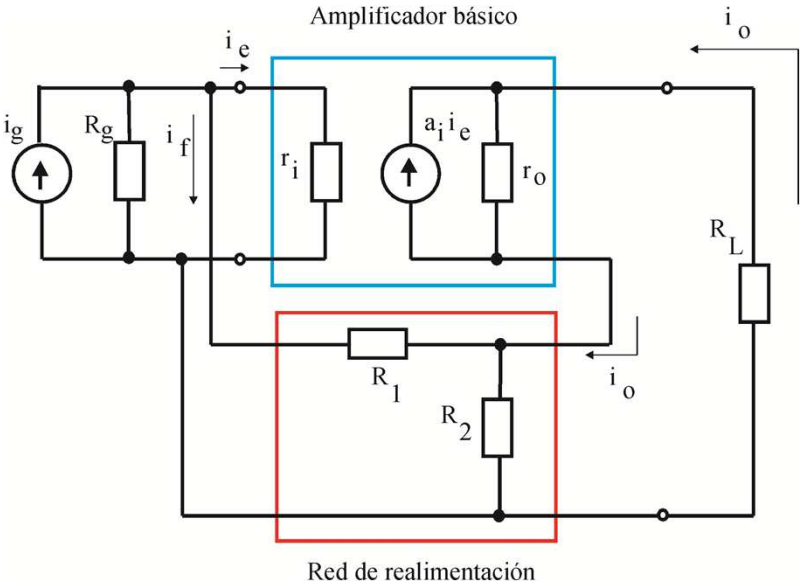


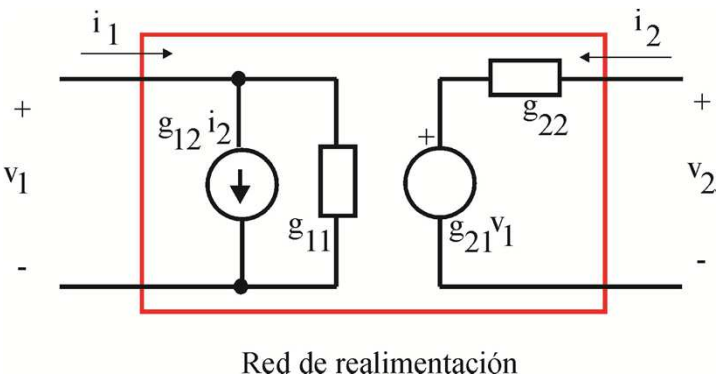


$$R_{if} = \frac{v_i}{i_i} = \frac{r_i}{(1 + a_i f)}$$

$$R_{of} = \frac{v_o}{i_o} = r_o(1 + a_i f)$$

Paralelo - Serie. Esquema real





$$g_{11} = \left.\frac{i_1}{v_1}\right|_{i_2=0} = \frac{1}{R_1 + R_2}$$

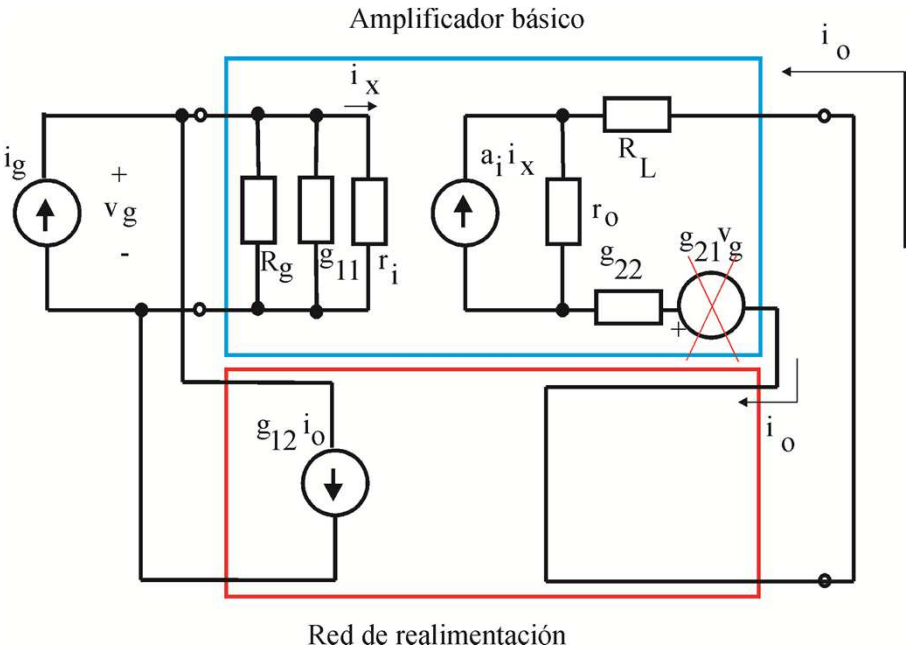
$$g_{12} = \left.\frac{i_1}{i_2}\right|_{v_1=0} = -\frac{R_2}{R_1 + R_2}$$

$$g_{21} = \left.\frac{v_2}{v_1}\right|_{i_2=0} = \frac{R_2}{R_1 + R_2}$$

$$g_{22} = \left.\frac{v_2}{i_2}\right|_{v_1=0} = \frac{R_1 R_2}{R_1 + R_2}$$

Paralelo - Serie. Esquema real

$$a = \frac{i_o}{i_g}(f = 0) = -\frac{r_o}{r_o + R_L + g_{22}} a_i \frac{R_g \parallel g_{11}{}^{-1}}{R_g \parallel g_{11}{}^{-1} + r_i}$$



$$f = g_{12} = -\frac{R_2}{R_1 + R_2}$$

$$\boxed{A = \frac{i_o}{i_g} = \frac{a}{1 + af}}$$

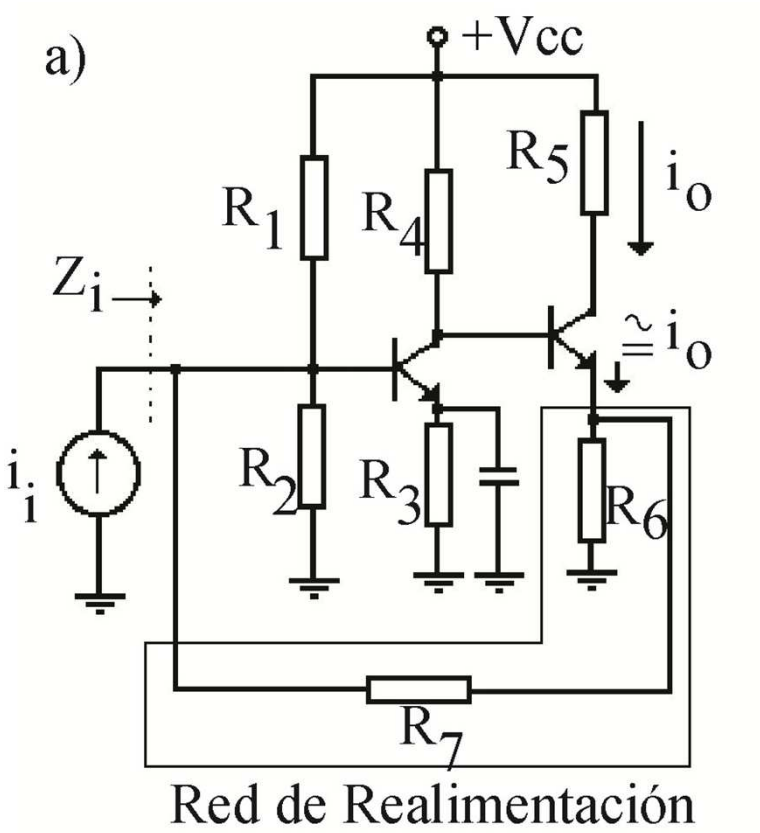
$$R_{io} = r_i \parallel R_g \parallel g_{11}{}^{-1}$$

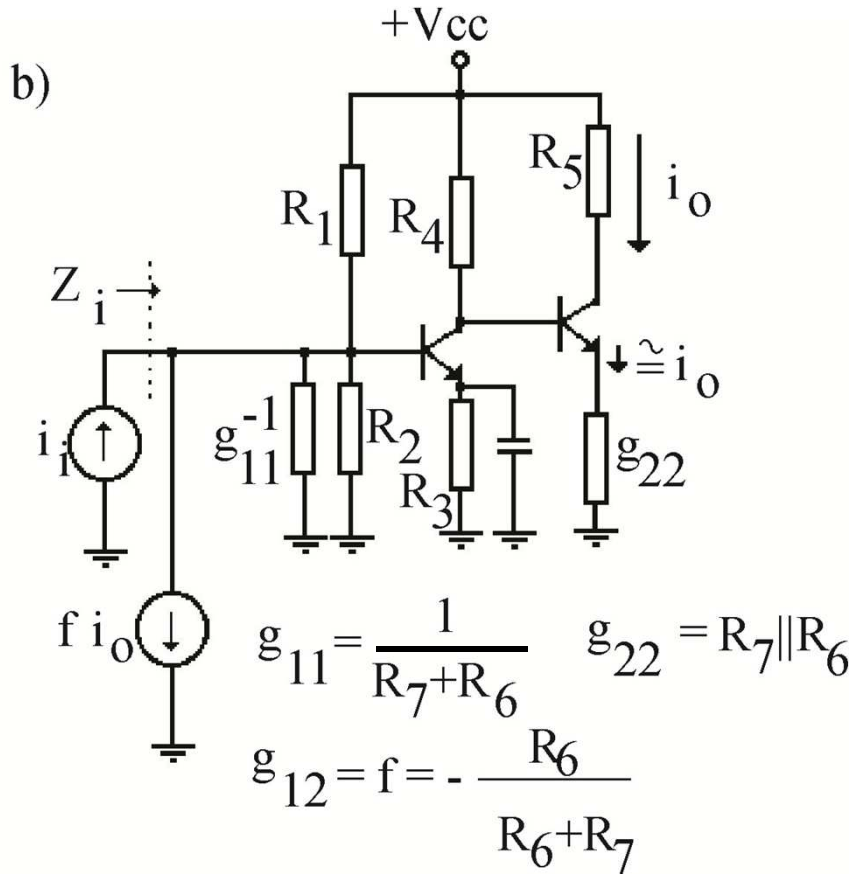
$$\boxed{R_{if} = \frac{R_{io}}{1 + af}}$$

$$R_{oo} = r_o + R_L + g_{22}$$

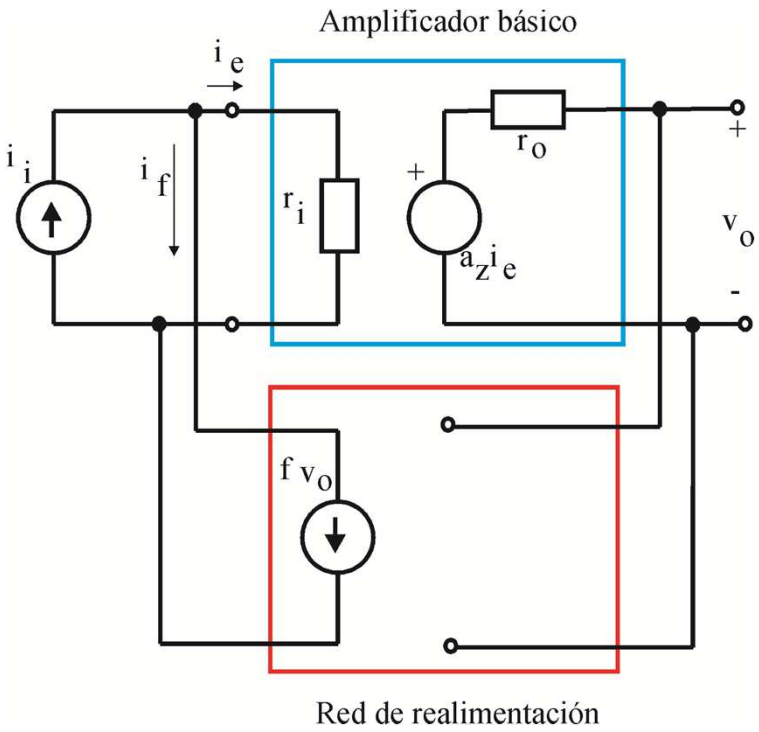
$$\boxed{R_{of} = R_{oo}(1 + af)}$$

Paralelo - Serie. Ejemplo



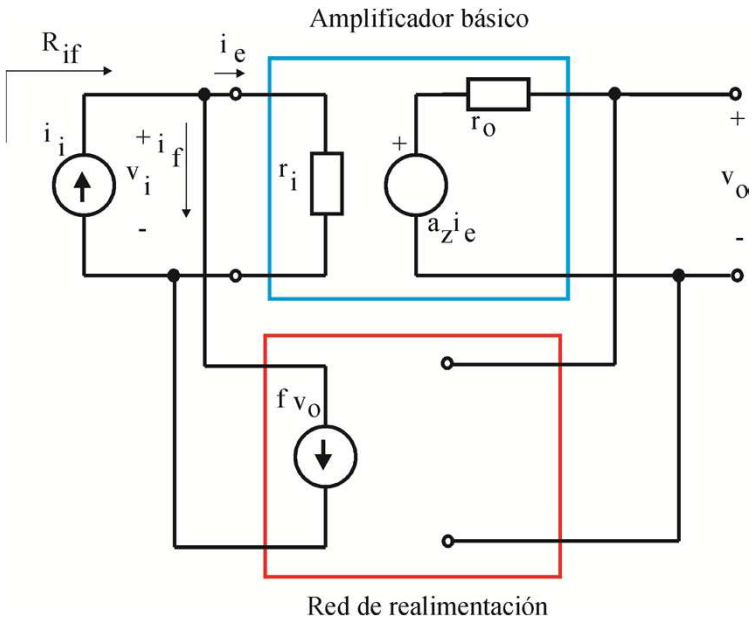


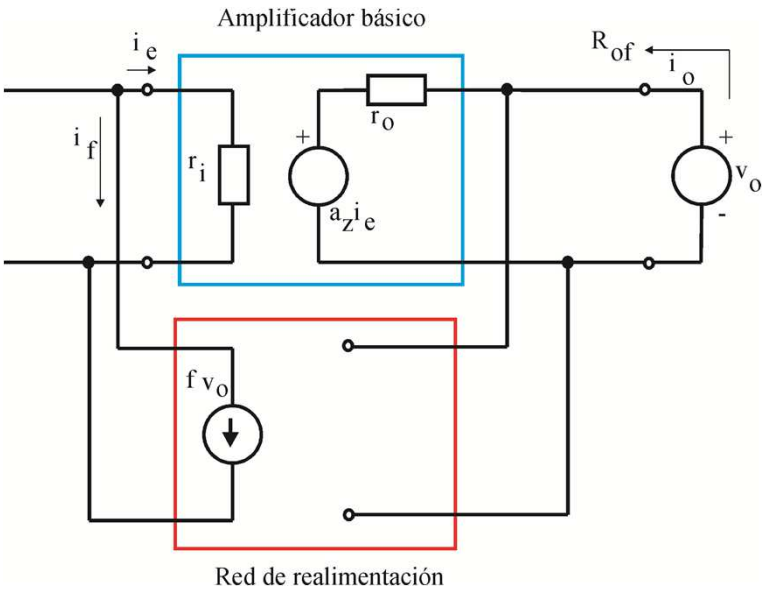
Paralelo - Paralelo. Esquema ideal



$$A_z = \frac{v_o}{i_i} = \frac{a_z}{(1 + a_z f)}$$

Paralelo - Paralelo. Esquema ideal

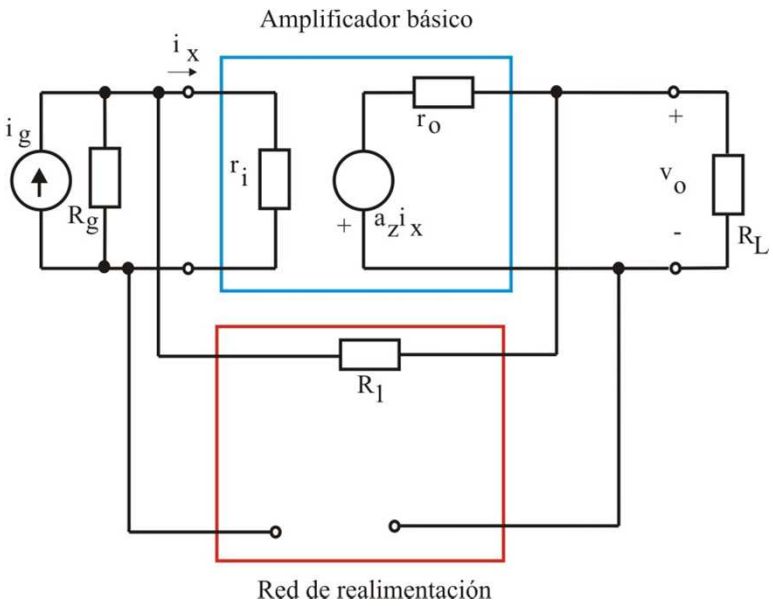


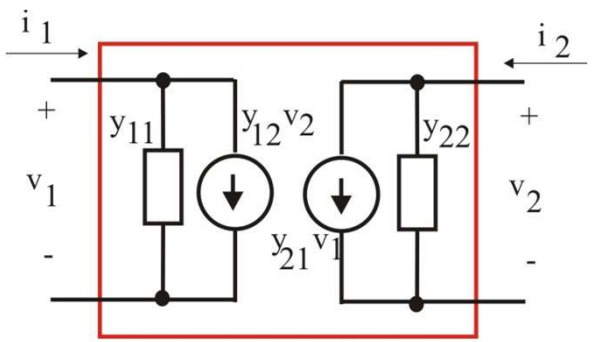


$$R_{if} = \frac{v_i}{i_i} = \frac{r_i}{(1 + a_z f)}$$

$$R_{of} = \frac{v_o}{i_o} = \frac{r_o}{(1 + a_z f)}$$

Paralelo - Paralelo. Esquema real





$$y_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0} = \frac{1}{R_1}$$

$$y_{12} = \left.\frac{i_1}{v_2}\right|_{v_1=0} = -\frac{1}{R_1}$$

$$y_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0} = -\frac{1}{R_1}$$

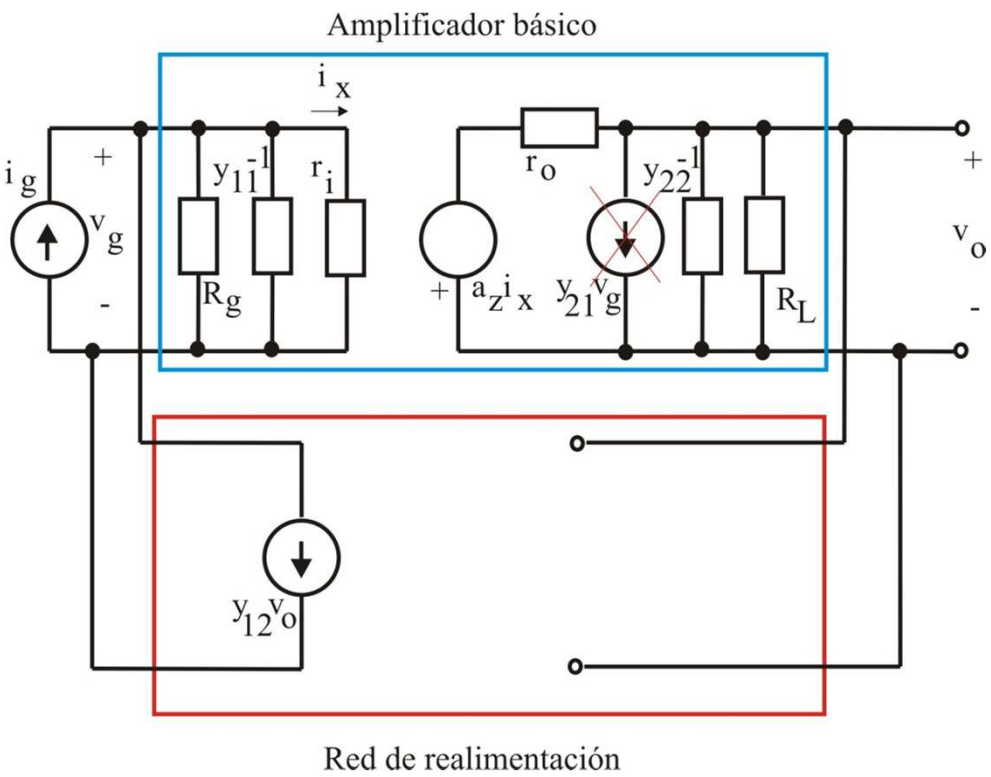
$$y_{22} = \left.\frac{i_2}{v_2}\right|_{v_1=0} = \frac{1}{R_1}$$

Paralelo - Paralelo. Esquema real

$$a = \frac{v_o}{i_g}(f=0) = -\frac{R_L||{y_{22}}^{-1}}{R_L||{y_{22}}^{-1} + r_o} a_z \frac{R_g||{y_{11}}^{-1}}{R_g||{y_{11}}^{-1} + r_i}$$

$$f = y_{12}$$

$$\boxed{A = \frac{a}{1+af}}$$



$$R_{io} = \frac{1}{\frac{1}{r_i} + \frac{1}{R_g} + y_{11}}$$

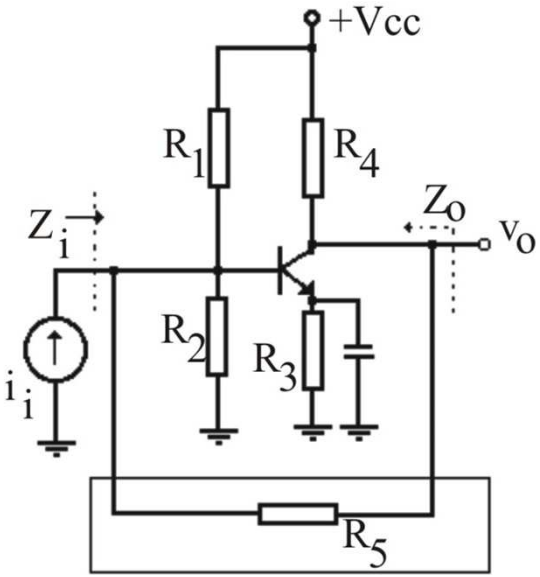
$$\boxed{R_{if} = \frac{R_{io}}{1+af}}$$

$$R_{oo} = \frac{1}{\frac{1}{R_L} + \frac{1}{r_o} + y_{22}}$$

$$\boxed{R_{of} = \frac{R_{oo}}{1+af}}$$

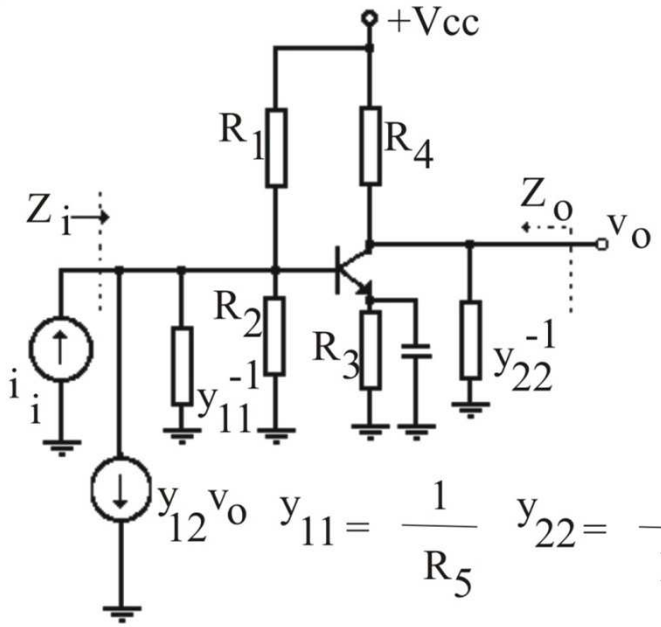
Paralelo - Paralelo. Ejemplo

a)



Red de Realimentación

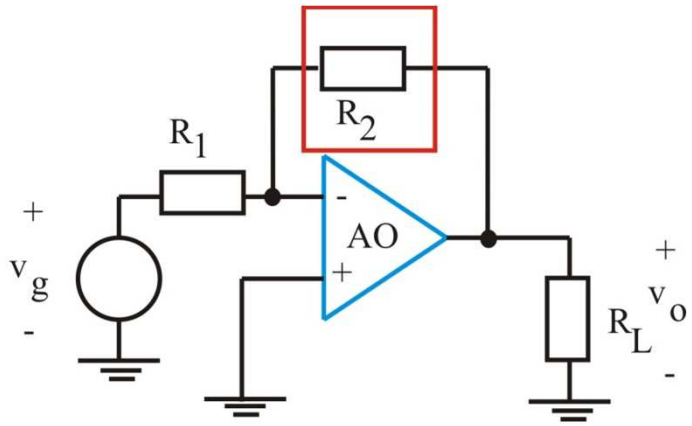
b)

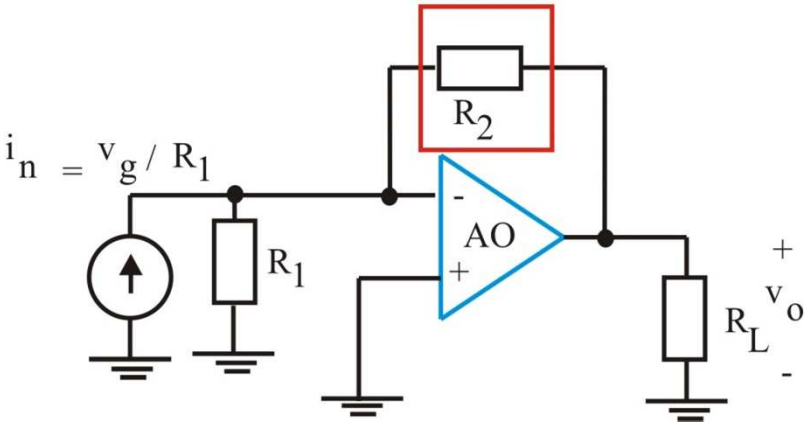


$$y_{11} = \frac{1}{R_5} \qquad y_{22} = \frac{1}{R_5}$$

$$y_{12} = f = -\frac{1}{R_5}$$

Paralelo - Paralelo. Ejemplo





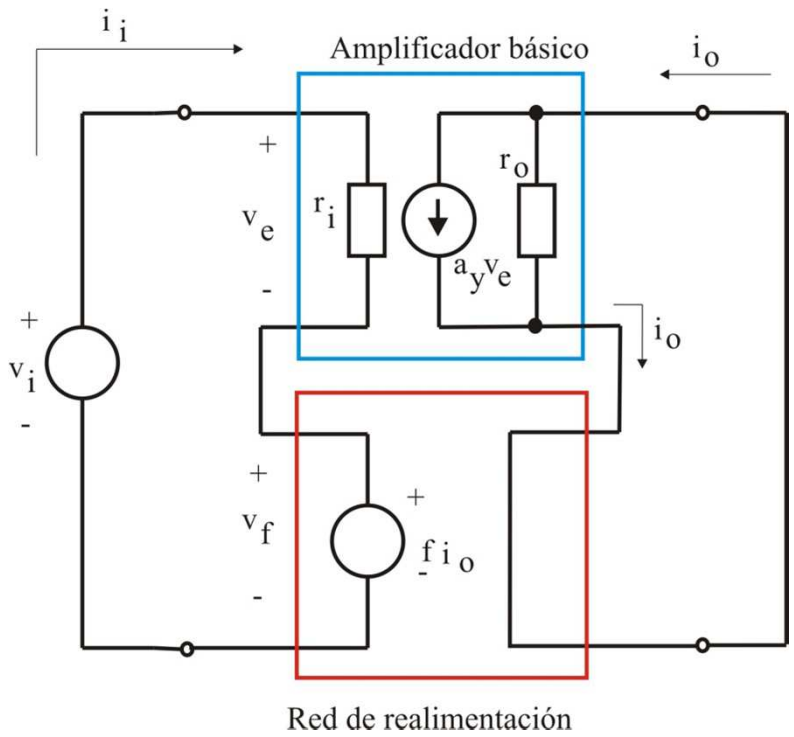
$$f = y_{12} = -\frac{1}{R_2}$$

Si af $\gg$ 1

$$A = \frac{v_o}{i_n} \cong \frac{1}{f} = -R_2$$

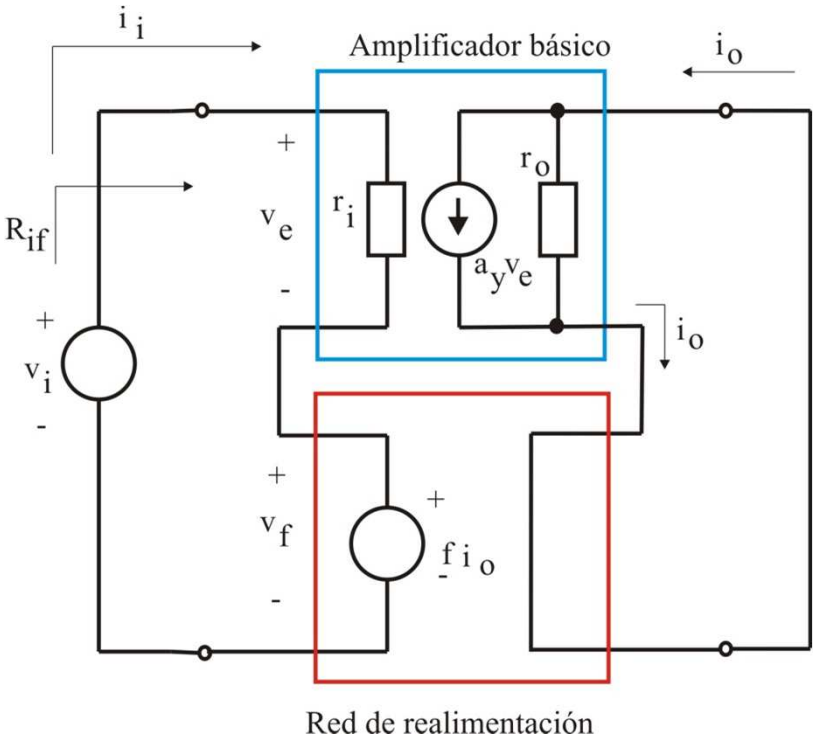
$$\boxed{\frac{v_o}{v_g} = \frac{v_o}{i_n}\frac{i_n}{v_g} = A\frac{1}{R_1} = -\frac{R_2}{R_1}}$$

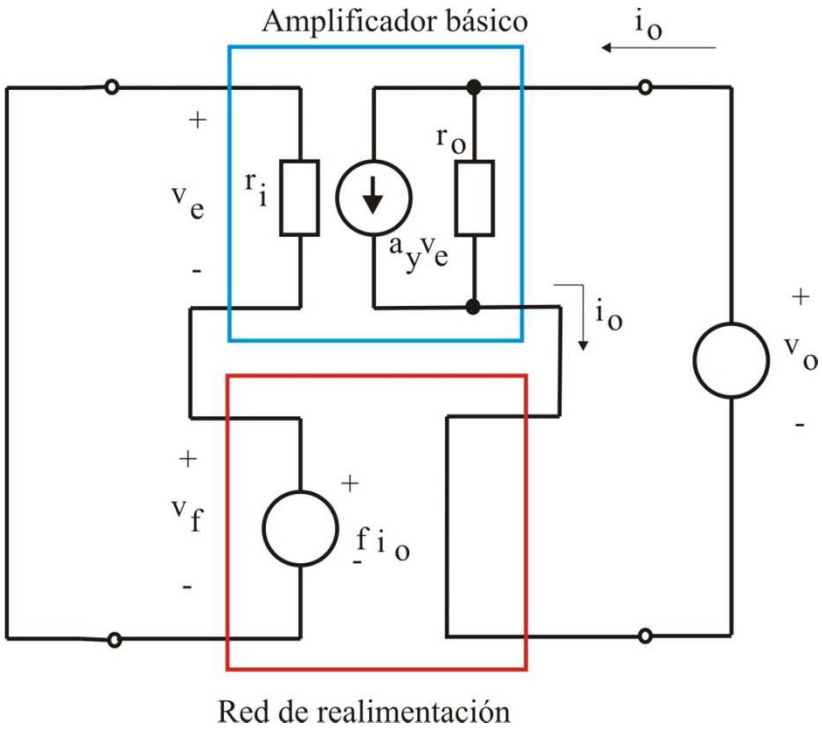
Serie - Serie. Esquema ideal



$$A_y = \frac{i_o}{v_i} = \frac{a_y}{(1 + a_y f)}$$

Serie - Serie. Esquema ideal

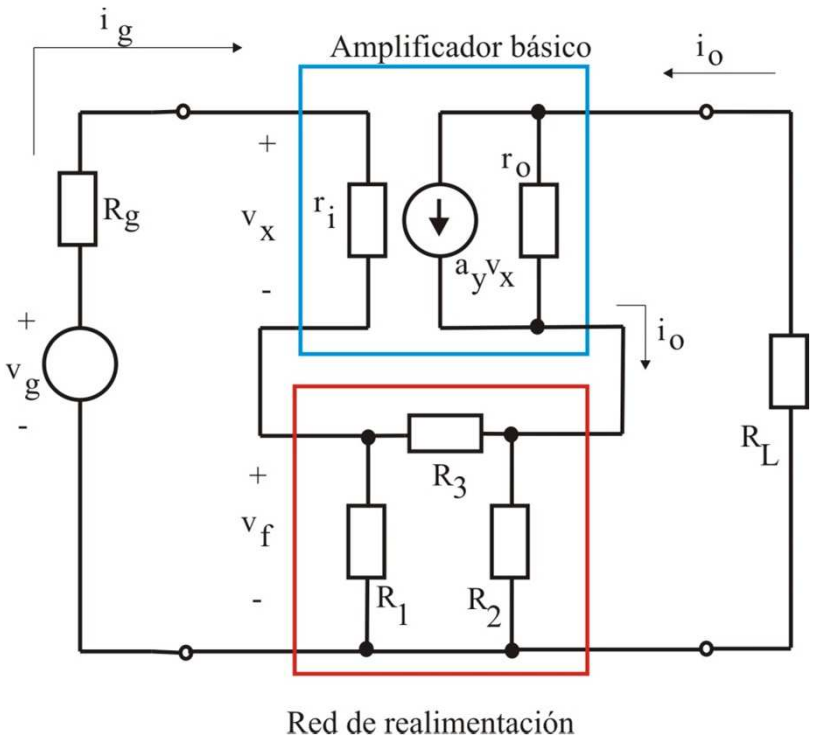


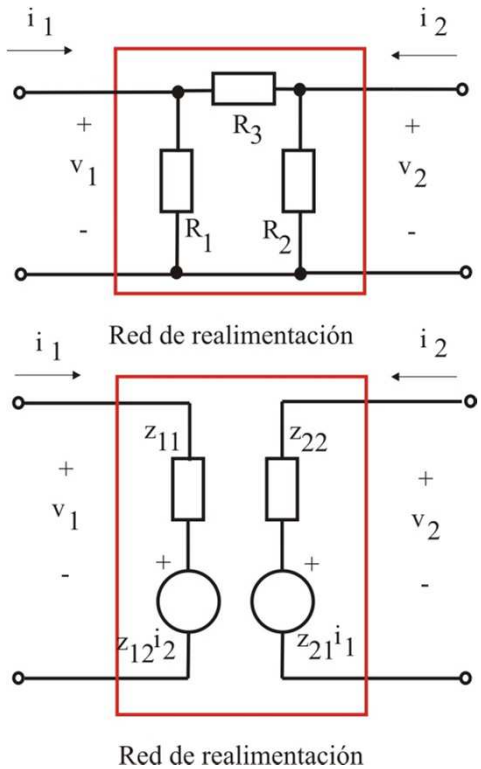


$$R_{if} = \frac{v_i}{i_i} = r_i(1 + a_y f)$$

$$R_{of} = \frac{v_o}{i_o} = r_o(1 + a_y f)$$

Serie - Serie. Esquema real





$$z_{11} = \left.\frac{v_1}{i_1}\right|_{i_2=0} = R_1||(R_2 + R_3)$$

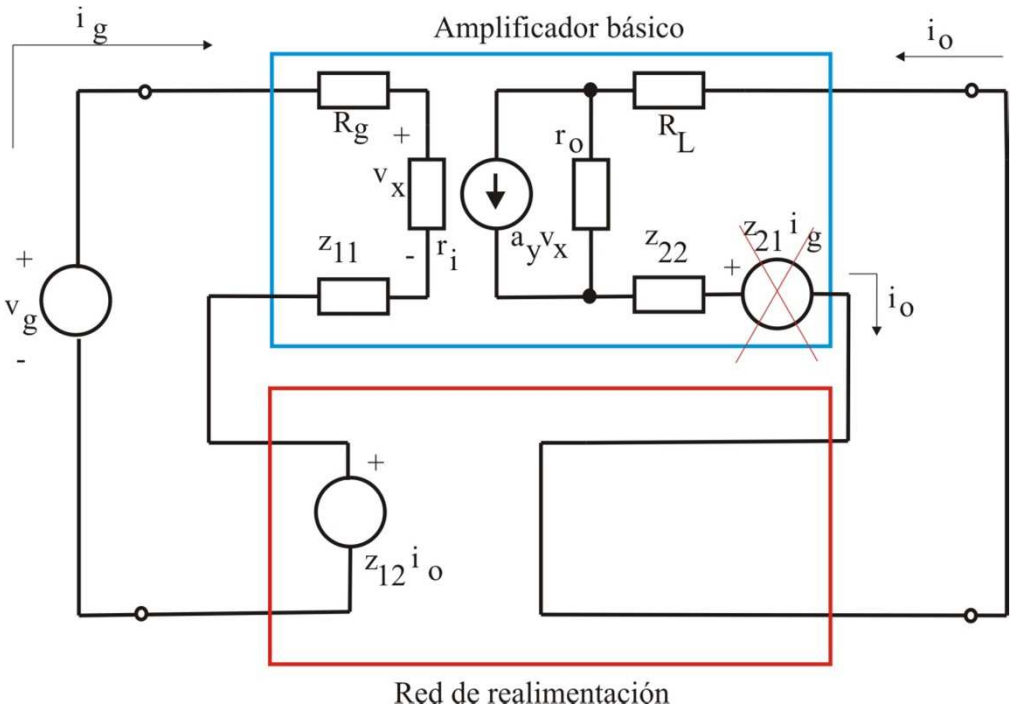
$$z_{12} = \left.\frac{v_1}{i_2}\right|_{i_1=0} = \frac{R_2}{R_1 + R_2 + R_3} R_1$$

$$z_{21} = \left.\frac{v_2}{i_1}\right|_{i_2=0} = \frac{R_1}{R_1 + R_2 + R_3} R_2$$

$$z_{22} = \left.\frac{v_2}{i_2}\right|_{i_1=0} = R_2||(R_1 + R_3)$$

Serie - Serie. Esquema real

$$a = \frac{i_o}{v_g}(f = 0) = \frac{r_o}{r_o + R_L + z_{22}} a_y \frac{r_i}{r_i + R_g + z_{11}}$$



$$f = z_{12} = \frac{R_2}{R_1 + R_2 + R_3} R_1$$

$$\boxed{A = \frac{i_o}{v_g} = \frac{a}{1 + af}}$$

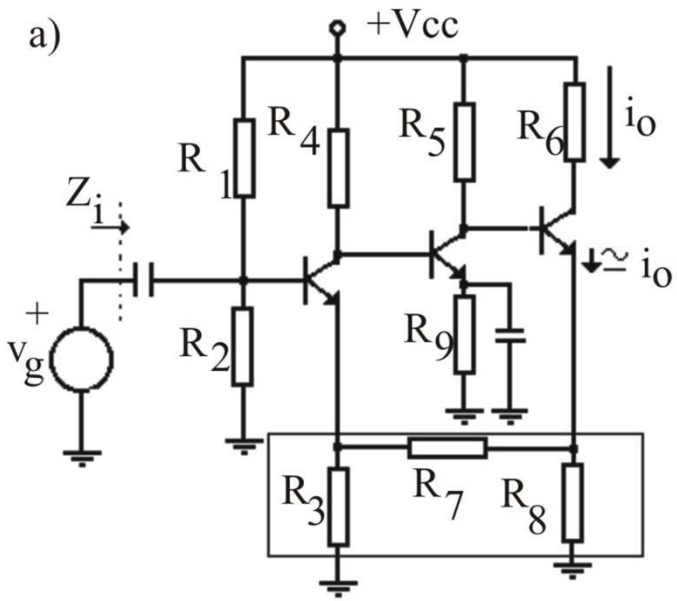
$$R_{io} = r_i + R_g + z_{11}$$

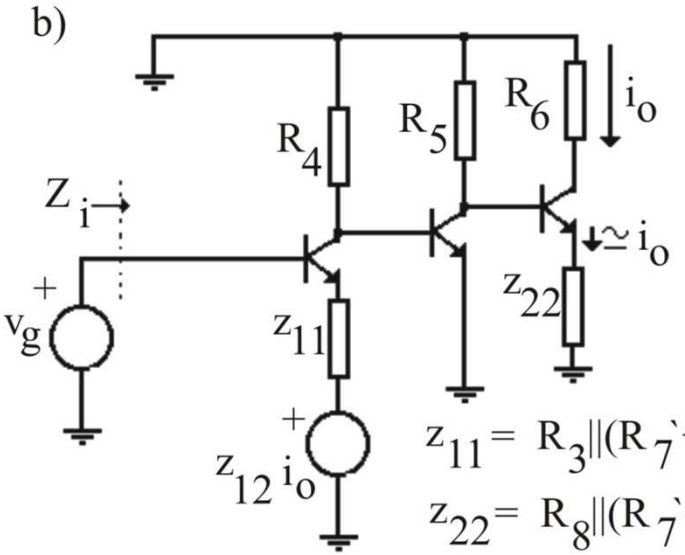
$$\boxed{R_{if} = R_{io}(1 + af)}$$

$$R_{oo} = r_o + R_L + z_{22}$$

$$\boxed{R_{of} = R_{oo}(1 + af)}$$

Serie - Serie. Ejemplo





$$z_{11} = R_3 \| (R_7' + R_8)$$

$$z_{22} = R_8 \| (R_7' + R_3)$$

$$z_{21} = f = \frac{R_3 R_8}{R_3 + R_7 + R_8}$$